

Guideline Title: Subarachnoid Hemorrhage, Non-Traumatic

Guideline Number: GUID-3203-M013

Issue date: 09/09/2014	Replaces Dept. Policy: Non-traumatic Subarachnoid Hemorrhage M013
Revision dates: 4/30/24	Developed by: Air Care Team
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Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To recognize the patient with non-traumatic subarachnoid hemorrhage (SAH) and to provide and maintain adequate oxygenation/ventilation and tissue perfusion. To provide emergent treatment, prevent secondary brain injury, and expeditious transport to the nearest appropriate medical facility.
- II. **DEFINITIONS:**
 - a. CPP – Cerebral perfusion pressure = MAP-ICP
 - b. GCS - Glasgow Coma Scale
 - c. ICP – Intracranial pressure
 - d. MAP – Mean arterial pressure
 - e. SBP – Systolic blood pressure
- III. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by *GUID3203-G002 – General Patient Care*.
 - b. Perform a complete and timely neurological exam.
 - c. Consider Ottawa Subarachnoid hemorrhage rule for scene suspected SAH: Age ≥ 40 yo, neck pain/stiffness, limited neck flexion on exam, witnessed loss of consciousness, onset during exertion, and thunderclap headache.
 - d. Maintain HOB at 30-degree angle if spinal cord injury is not suspected.
 - e. Treat pain and/or anxiety per *GUID3203-G005 - Analgesia and Sedation Management*.
 - f. Treat nausea/vomiting per *GUID3203-G004 - Motion Sickness/Nausea*.
 - g. In the intubated patient, short acting agents such as propofol are ideal.
 - h. Target SBP < 140 mmHg unless otherwise directed by the receiving physician. Preferred medications:
 - i. **Nicardipine infusion at 5mg/h.** Increase by 2.5mg/h IV every 5-15 minutes to maximum of 15mg/h.
 - ii. **Esmolol bolus 500 mcg/kg/min IV** over 1 minute followed by **Esmolol infusion at 50 mcg/kg/min.** Titrate every 5 minutes by repeating 500 mcg/kg/min bolus over 1 minute and then increasing infusion by 50 mcg/kg/min. Maximum dose 300 mcg/kg/min
 - iii. **Labetalol 10-20 mg IV** over 2 minutes. May repeat every 10 minutes until goal blood pressure obtained or maximum total dose of 300mg. Avoid if HR < 60 bpm.

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- i. Treat Hypotension aggressively. If MAP < 70mmHg with mental status changes, refer to per *GUID3203-M011 - Hypotension*.
- j. If patient taking anticoagulants, consider reversal agents.
 - i. Attempt to determine timing of last anticoagulant dose
 - ii. Obtain copies of appropriate baseline coagulation studies such as: PT, aPTT, INR, TEG, anti-Xa
 - iii. Continue appropriately ordered reversal agent for patients on anticoagulant medications (anti-inhibitor coagulant complex [FEIBA], idarucizumab [Praxbind], Vitamin K, coagulation factor Xa recombinant [Andexanet alpha]) per transferring or receiving facility orders.
 - iv. For patients taking Warfarin with an INR > 1.4, consider with sending physician
 - 1. **Vitamin K 1-2.5 mg PO or 10 mg IV** if no prosthetic heart valve
 - 2. **FEIBA 1000 units/20 ml IV** over 15 minutes
 - 3. **Fresh frozen plasma 2 units IV** over 15 minutes
 - v. For patients taking oral factor Xa Inhibitors such as Apixaban (Eliquis), Edoxaban (Sayvaysa), or Rivaroxaban (Xarelto), consider with sending physician
 - 1. **FEIBA 2000 units/40 ml IV** over 15 minutes
 - 2. **Tranexamic acid 1 gram IV** over 10 minutes
 - vi. If FEIBA was ordered at sending facility:
 - 1. Wait for the FEIBA medication before transferring the patient. If the patient is at the point of herniating, call the receiving physician to discuss if they would prefer you to transfer without delay.
 - 2. The IV access used for FEIBA is dedicated to only FEIBA for time of administration.
 - 3. Ensure the MD/RN at receiving facility is aware of the time that FEIBA was given.
 - 4. Observe for infusion reactions, thromboembolic events, & allergic reactions.
 - vii. For patients taking Dabigatran (Pradaxa), consider with sending physician
 - 1. **Idarucizumab 2.5 grams IV** every 10 minutes x 2 doses
- k. For interfacility transfers only and proven by CT scan: If signs of cerebral herniation caused by increased intracranial pressure from KNOWN intracranial hemorrhage
 - i. Signs and symptoms include unilateral dilated pupil, nonreactive pupil, decreased in GCS of 2 or more in a patient with a GCS less than 9, and/or spontaneous posturing, CT scan showing herniation or significant midline shift.
 - ii. Assess for other causes of hypoxia, hypotension, or over-sedation.
 - iii. Hyperventilate to an ETCO₂ of 30-35 mmHg
 - iv. Administer **Hypertonic 3% Saline 500 ml IV** over 10 minutes for adult patients.
 - v. If a sending facility prefers, consider obtaining and administering **Mannitol 1 gram/kg IV** over 5-10 minutes using a filter on IV tubing.
- l. If seizures are present, treat per *GUID3203-M015 - Seizures*.
- m. For confirmed intracranial hemorrhage in the immediate post-hemorrhagic phase, administer **Levetiracetam 1 gram IV** over 2-5 minutes as a prophylactic anticonvulsant
- n. Document Nimodipine dose if previously received from sending facility.

IV. DOCUMENTATION:

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a. EMS Charts

V. **REFERENCES:**